

Supplementary methods and results

Companion to: When does decision-focused learning add value to residential battery control? Private costs and AC feeder consequences.

S1. Source chronology, coverage and sample formation

The source-clock half-hours and data-quality indicators are retained. Nominal PV capacity is taken from information available before each target day, not estimated by outcome maximization. Known-household training has 132,200 days and validation 44,373; test evaluation has 88,609 known and 20,032 held-out GC/PV-valid days. The CL-valid absolute-cost sample is 107,713 days. Estimated measurements and required-channel gaps are excluded, and no missing meter record is synthesized. Feeder complete dates must satisfy every assigned household's required channels. The expanded private comparison and joint feeder comparison thus answer different estimands.

Table S1a. Jointly clean dates by calendar month.

Month	Calendar	Panel 1	Panel 2	Panel 3	Held-out
2012-07	31	0	12	31	31
2012-08	31	8	0	31	23
2012-09	30	20	0	30	30
2012-10	31	31	0	31	31
2012-11	30	30	1	30	30
2012-12	31	31	31	31	31
2013-01	31	31	25	25	25
2013-02	28	7	0	0	0
2013-03	31	0	0	0	0
2013-04	30	17	10	18	21
2013-05	31	23	20	22	22
2013-06	30	21	30	27	30
Total	365	219	129	276	274

Counts are actual evaluated dates, not bootstrap replicates. Panel 2 coverage is 35.3% of the calendar. Zero months remain in the calendar-axis resampling array.

Table S1b. Load and PV on retained feeder dates versus individually valid dates.

Panel	Joint dates	Joint demand (kWh/day)	Joint PV (kWh/day)	Individual days	Individual demand	Individual PV
1	219	18.24	5.91	19968	18.98	5.70
2	129	18.41	6.05	19764	17.13	6.94
3	276	17.85	5.86	19897	17.43	5.86
Held-out	274	18.22	5.65	19943	17.53	5.75

Both descriptions use observed gross GC+CL and PV. Individually valid days need not support all-site AC replay and are not a missing-data correction. Main private GC/PV comparisons additionally allow CL cancellation.

S2. Learning and baseline specifications

The 774-input PV network uses 128/64 ReLU hidden units and 48 softplus outputs. Supervised configuration selection precedes three-seed initialization. DFL and continued MSE share 4,096 training

pairs, 1,024 validation pairs, batch 32, four epochs, AdamW weight decay 10^{-4} , gradient clipping at norm 1, and learning-rate candidates 10^{-4} and 3×10^{-5} . Initial checkpoints are eligible; downstream validation cost chooses each arm's checkpoint. All initialization, sampling and selection records are retained, including unsuccessful settings. Night restriction fixes 00:00–04:00 and 21:00–24:00 to zero. The window follows the measured generation support of the Ausgrid test records.

Three-offset and twelve-offset calibration use bounds $0.25 \times \tanh(\text{parameter})$ in capacity-normalized units, three epochs of Adam at 0.03 and training batch 32. Calibration uses 2,048 known training and 2,048 validation examples. Validation chooses shrinkage in $\{0, 0.25, 0.5, 0.75, 1\}$. The seasonal variant crosses three fixed daytime windows with DJF/MAM/JJA/SON. Direct net-load prediction uses signed linear outputs, the same inputs, and full training/validation splits; two prespecified AdamW configurations are selected by validation MSE with at most 30 epochs and patience six. Six four-hour windows by four seasons give 24 cost-calibration parameters. The primary direct-net comparison applies an explicit support-matched nighttime projection, and the unrestricted signed variant is reported separately.

The stronger demand network has 256/128 hidden units and softplus outputs. Validation chooses the mixture of the original HGB and this network among weights 0, 0.25, 0.5, 0.75, 1; it selects 0.5. A previous-day profile is the persistence comparator. Residual scenarios use the original frozen forecast backbone and 20 joint load/PV daily residual paths from 147 clean validation dates. A deterministic donor policy supplies household-held-out residuals. Shared battery actions apply in every scenario; no oracle recourse, network coordination or test residual fitting is introduced.

Table S2. Complete all-household baseline comparison.

Tariff	Method	Full cost	Retail	Throughput	Paired Δ (cents)	95% interval
tou	msecont	3.53815	3.29425	12.195	0	Reference
tou	dfl	3.52683	3.28929	11.877	-1.131	-1.282 to -0.997
tou	calibrated3	3.53202	3.29433	11.884	-0.612	-0.685 to -0.537
tou	stochastic20	3.50217	3.28616	10.801	-3.593	-3.899 to -3.277
tou	seasonal12	3.53235	3.29400	11.918	-0.579	-0.723 to -0.432
tou	net mse night	3.57396	3.32495	12.451	3.568	3.151 to 4.028
tou	net calibrated24 night	3.55907	3.32795	11.556	2.087	1.733 to 2.485
tou	net mse	3.57398	3.32496	12.451	3.570	3.153 to 4.030
tou	net calibrated24	3.56029	3.32817	11.606	2.209	1.847 to 2.621
flat	msecont	3.74045	3.67408	3.318	0	Reference
flat	dfl	3.72835	3.65093	3.871	-1.216	-1.426 to -0.987
flat	calibrated3	3.73478	3.66022	3.728	-0.565	-0.703 to -0.408
flat	stochastic20	3.69327	3.60592	4.367	-4.725	-5.187 to -4.177

Tariff	Method	Full cost	Retail	Throughput	Paired Δ (cents)	95% interval
flat	seasonal12	3.73705	3.66160	3.773	-0.339	-0.526 to -0.138
flat	net mse night	3.75486	3.69202	3.142	1.443	1.153 to 1.769
flat	net calibrated24 night	3.74349	3.67416	3.466	0.306	0.045 to 0.602
flat	net mse	3.75502	3.69214	3.144	1.458	1.170 to 1.785
flat	net calibrated24	3.74384	3.67369	3.507	0.341	0.078 to 0.630

Cost and retail: AUD/day, same 107,713 CL-valid days; listed throughput is independently reaggregated on that same mask. Paired Δ uses 108,641 GC/PV days, identical CL charge cancels. Negative Δ favors the row method over continued MSE. Seed 11; unsuccessful candidates are retained.

Predictive metrics are diagnostic, not downstream selection criteria for task learning. The common and selected mixed demand validation RMSEs are 0.547316 and 0.542329 kW, respectively; the neural-only demand model has 0.5510 kW validation RMSE. On the 244 known households and 88,609 test household-days, the original unrestricted signed direct-net MSE has about 0.6253 kW test RMSE versus about 0.6184 kW for the separated MSE net input. These diagnostic values come from the unrestricted signed variant rather than the support-matched baseline and do not establish architecture necessity. Complete split and role metrics accompany the machine-readable reproduction records.

S3. Resource conditions and numerical definitions

Resource timings use 1,024 fixed validation pairs, 20 warmups and three serial method repeats, with one numerical thread per subprocess. Other analysis jobs run concurrently. Windows CPU-clock granularity affects individual process-CPU measurements; both wall and CPU summaries are retained. Common original supervised fitting is excluded. Continued-MSE/DFL resource refits reproduce their original fitting protocol, while performance uses the original frozen models. All sample counts, validation procedures and memory scope are reported rather than treating different fitting procedures as equal-budget attribution tests.

Table S3. Full resource measurement.

Method	Learn (s)	Mean wall (ms)	p95 wall (ms)	Mean CPU (ms)	Replay RSS (MB)	Learn RSS (MB)
msecont	21.76	1.453	1.883	1.429	618.4	854.8
dfl	385.58	1.461	1.915	1.419	619.2	938.8
calibrated3	87.31	1.476	1.889	1.460	807.9	936.2
stochastic20	0.01	9.344	11.754	9.145	618.5	Undefined

RSS includes interpreter, dependencies and loaded arrays. No memory claim for a hypothetical embedded deployment is inferred. SAA learning is residual extraction only.

S4. Direction, severity, worst locations and event persistence

Table S4a. Voltage direction and conditional severity.

Network	Source	Method	Over (%)	Under (%)	Cond. over (pu)	Cond. under (pu)
eulv	1.050	dfl	17.901	0.023	0.003159	0.006814
eulv	1.050	msecont	18.751	0.026	0.003144	0.007081

Network	Source	Method	Over (%)	Under (%)	Cond. over (pu)	Cond. under (pu)
eulv	1.025	df1	0.000	0.198	Undefined	0.013400
eulv	1.000	df1	0.000	1.359	Undefined	0.013459
rural1	1.025	df1	0.000	0.000	Undefined	Undefined
semiurb4	1.025	df1	0.000	0.000	Undefined	Undefined
eulv	1.025	msecont	0.000	0.207	Undefined	0.013702
eulv	1.000	msecont	0.000	1.460	Undefined	0.013346
rural1	1.025	msecont	0.000	0.000	Undefined	Undefined
semiurb4	1.025	msecont	0.000	0.000	Undefined	Undefined

All comparisons use panel 1, seed 11, full ToU. Conditional magnitude is pooled over event-bearing customer half-hours; no event means undefined, not zero.

Table S4b. Voltage tails, worst sites and persistence.

Network	Source	Method	q99 (pu)	Max (pu)	Worst O/U site	Longest O (h)	Longest U (h)
eulv	1.050	df1	1.058259	1.071583	53/29	14.0	0.5
eulv	1.050	msecont	1.058341	1.071964	50/29	14.5	0.5
eulv	1.025	df1	1.033462	1.047158	Undefined/29	0.0	2.0
eulv	1.000	df1	1.008671	1.022767	Undefined/29	0.0	7.0
rural1	1.025	df1	1.024385	1.029856	Undefined/Undefined	0.0	0.0
semiurb4	1.025	df1	1.024468	1.028303	Undefined/Undefined	0.0	0.0
eulv	1.025	msecont	1.033545	1.047548	Undefined/29	0.0	2.0
eulv	1.000	msecont	1.008758	1.023165	Undefined/29	0.0	7.0
rural1	1.025	msecont	1.024355	1.028839	Undefined/Undefined	0.0	0.0
semiurb4	1.025	msecont	1.024454	1.027830	Undefined/Undefined	0.0	0.0

Site indices are 1-based household assignment slots. In SimBench, these 55 slots repeat the voltage of the corresponding 13 or 41 native load locations, and pooled voltage statistics weight those 55 assignments; they are not native bus indices. No-event worst sites are undefined. Episode lengths concatenate contiguous observed half-hours and break on missing dates; they describe half-hour-average excursions, not subinterval dynamics. Node quantiles, episode counts and p95 lengths remain in machine arrays.

S5. Physical and mapping brackets

Table S5. Matched network-model sensitivity.

Family	Setting	Voltage Δ (pp)	Import Δ (kW)	Export Δ (kW)	Loss Δ (kWh/day)
physical brackets	PF 0.95; Z 0.9	-0.847	-1.758	0.387	-0.374
physical brackets	PF 0.9; Z 1.0	-0.734	-1.776	0.387	-0.422
physical brackets	PF 1.0; Z 1.0	-1.083	-1.771	0.387	-0.413
physical brackets	PF 0.95; Z 1.1	-0.876	-1.791	0.387	-0.465

Family	Setting	Voltage Δ (pp)	Import Δ (kW)	Export Δ (kW)	Loss Δ (kWh/day)
mapping	0	-0.965	-1.765	0.386	-0.445
mapping	1	-0.667	-1.752	0.386	-0.382
mapping	2	-0.953	-1.755	0.385	-0.468
mapping	3	-0.735	-1.779	0.386	-0.446
mapping	4	-0.886	-1.754	0.386	-0.431
mapping	5	-0.823	-1.776	0.388	-0.455
mapping	6	-0.716	-1.751	0.385	-0.395
mapping	7	-0.795	-1.750	0.387	-0.427
mapping	8	-0.837	-1.746	0.389	-0.425
mapping	9	-0.812	-1.745	0.386	-0.381

DFL minus continued MSE; seed 11, panel 1. Transformer rating and parameters are unchanged during line-impedance scaling. These are bounded model sensitivity cases, not calibration to an observed physical feeder.

S6. Frozen transport, matched retraining and inventory accounting

Table S6. Complete private-cost condition matrix.

Family	Setting	Question	Paired days	Δ (cents/day)	95% interval	Benefiting HH (%)
Load	persistence	frozen	108641	0.312	-0.064 to 0.673	55.5
Load	persistence	matched	108641	-1.475	-1.847 to -1.138	97.7
Load	strong	frozen	108641	-1.194	-1.343 to -1.057	96.7
Load	strong	matched	108641	-1.235	-1.417 to -1.059	96.7
Wear	0	frozen actions repriced	108641	-0.495	-0.660 to -0.346	86.0
Wear	0	frozen signals reoptimized	108641	-0.495	-0.660 to -0.346	86.0
Wear	0	matched signals reoptimized	108641	-0.792	-0.987 to -0.605	94.3
Wear	0.01	frozen actions repriced	108641	-0.813	-0.969 to -0.675	94.6
Wear	0.01	frozen signals reoptimized	108641	-0.813	-0.969 to -0.675	94.6
Wear	0.01	matched signals reoptimized	108641	-0.868	-1.038 to -0.714	96.0
Wear	0.04	frozen actions repriced	108641	-1.767	-1.910 to -1.634	97.0
Wear	0.04	frozen signals reoptimized	108641	-1.767	-1.910 to -1.634	97.0

Family	Setting	Question	Paired days	Δ (cents/day)	95% interval	Benefiting HH (%)
Wear	0.04	matched signals reoptimized	108641	-2.416	-2.566 to -2.265	95.7
Continuous SOC	raw operating	frozen	108641	-1.135	-1.282 to -1.003	96.7
Continuous SOC	inventory adjusted	frozen	108641	-1.135	-1.283 to -1.001	96.7
Continuous SOC	raw operating	matched	108641	-1.313	-1.455 to -1.174	96.3
Continuous SOC	inventory adjusted	matched	108641	-1.313	-1.457 to -1.172	96.3

DFL minus its condition-matched MSE comparator, seed 11. Frozen repricing changes no action; frozen reoptimization changes the controller objective; matched retraining changes the signal under that objective. Macro estimands and held-out/known cohorts are separately retained in control_transfer_summary. Inventory-adjusted cost is not actual retail settlement.

Continuous SOC uses daily 24-hour QPs and two-day training rollout. Of the original sampled anchors, 4,088 training and 1,021 validation windows have the next day in the appropriate split. Each training window starts at 5 kWh; test carries energy across 328 complete evaluated household segments and does not reset every two days. The terminal inventory function includes both linear and quadratic terms. Segment corrections telescope as $V(\text{start}) - V(\text{end})$; invalid observation boundaries are common across arms. Average daily terminal energy is about 5.22 kWh for matched learners. The finite-difference gradient check includes 20 real examples, three joint net/SOC directions and three step scales, with a maximum best-scale absolute discrepancy of 5.32×10^{-10} . This establishes local KKT consistency, not a globally smooth active-set map.

S7. Network consequences after input and controller change

Table S7. Control-condition private and AC outcome contrasts.

Condition	Panel	Cost Δ (cents)	Over Δ (pp)	Over 95% interval	Import Δ (kW)	Loss Δ (kWh)
continuous frozen transfer	1	-1.153	-0.831	-0.975 to -0.700	-1.637	-0.400
continuous frozen transfer	Held-out	-1.030	-0.724	-0.922 to -0.537	-1.118	-0.406
continuous matched retraining	1	-1.377	-1.136	-1.310 to -0.972	-2.638	-0.564
continuous matched retraining	Held-out	-1.099	-1.040	-1.273 to -0.824	-2.082	-0.601
load persistence frozen transfer	1	0.303	-0.709	-0.845 to -0.572	-2.183	-0.409
load persistence frozen transfer	Held-out	0.479	-0.678	-0.831 to -0.534	-1.954	-0.447
load persistence matched retraining	1	-1.625	0.373	0.177 to 0.576	2.508	0.337
load persistence matched retraining	Held-out	-1.580	0.606	0.465 to 0.750	2.513	0.426
load strong frozen transfer	1	-1.220	-0.893	-1.049 to -0.743	-2.000	-0.435

Condition	Panel	Cost Δ (cents)	Over Δ (pp)	Over 95% interval	Import Δ (kW)	Loss Δ (kWh)
load strong frozen transfer	Held-out	-1.105	-0.772	-0.969 to -0.591	-1.293	-0.430
load strong matched retraining	1	-1.286	-0.759	-0.973 to -0.563	-1.185	-0.364
load strong matched retraining	Held-out	-1.123	-0.703	-0.946 to -0.473	-0.464	-0.382
wear0.01 frozen transfer	1	-0.818	-0.850	-0.994 to -0.714	-1.774	-0.419
wear0.01 frozen transfer	Held-out	-0.718	-0.746	-0.938 to -0.570	-1.228	-0.419
wear0.01 matched retraining	1	-0.825	-0.797	-0.960 to -0.627	-0.938	-0.328
wear0.01 matched retraining	Held-out	-0.771	-0.632	-0.838 to -0.445	-0.437	-0.332
wear0.04 frozen transfer	1	-1.735	-0.850	-0.994 to -0.714	-1.774	-0.419
wear0.04 frozen transfer	Held-out	-1.641	-0.746	-0.938 to -0.570	-1.228	-0.419
wear0.04 matched retraining	1	-2.385	-1.382	-1.561 to -1.228	-4.796	-0.830
wear0.04 matched retraining	Held-out	-2.179	-1.363	-1.597 to -1.155	-4.255	-0.886
wear0 frozen transfer	1	-0.512	-0.850	-0.994 to -0.714	-1.774	-0.419
wear0 frozen transfer	Held-out	-0.410	-0.746	-0.938 to -0.570	-1.228	-0.419
wear0 matched retraining	1	-0.743	-0.459	-0.667 to -0.254	0.606	-0.112
wear0 matched retraining	Held-out	-0.672	-0.365	-0.601 to -0.130	1.016	-0.149

Seed 11, full ToU, original European LV source 1.05 pu. Each group has its own matched reference; raw network private costs are not inventory-adjusted continuous economics. Panel numbering follows the main article. The matched-persistence counterexample is retained even though network outcomes worsen.

S8. Export protection and system rating

Table S8. Equal export-limit protection layer.

Panel	Cost Δ (cents/day)	Over Δ (pp)	Under Δ (pp)	Import Δ (kW)	Export Δ (kW)
1	-1.123	-0.850	-0.003	-1.774	0.387
2	-1.267	-1.254	-0.001	-1.415	0.155
3	-1.125	-1.044	-0.003	-1.452	0.135

DFL minus continued MSE, seed 11; all three fixed panels. Actual PV curtailment and lost export income are audited per household and half-hour. Official 3 kW guidance is prospective, not historical compliance. Battery actions are frozen before this common protection layer.

Curtailment is highly panel dependent. Panel 1 is almost nonbinding, Panel 2 curtails approximately 10.68–10.89 kWh/feeder-day, and Panel 3 approximately 0.090–0.115. Power-factor brackets are engineering assumptions supported by relevant operator guidance; they are not derived from these historical household meters. The main 800 kVA transformer is unchanged. In the rural 160 kVA topology, 55 historical households are aggregated to 13 native load locations, causing some nominal-rating exceedance even without voltage events; this is a constructed load-transfer stress case, not a measurement of that real circuit.

S9. Geography and household distribution checks

Table S9. Matched 27-household geographic-cohort check.

Outcome	Grouped gap	Dispersed gap	Difference of gaps	95% interval
daily operating cost aud	-0.012901	-0.012200	-0.000701	-0.002856 to 0.001178
daily over pct	-0.482735	-0.629823	0.147087	-0.024193 to 0.309537
daily under pct	0.000000	0.000000	0.000000	0.000000 to 0.000000
daily import peak kw	-0.373586	-0.591451	0.217866	0.075067 to 0.366068
daily export peak kw	0.081268	0.003852	0.077416	-0.026673 to 0.178800
daily loss kwh	-0.058477	-0.055584	-0.002893	-0.010446 to 0.004785

Gap is DFL minus continued MSE. Same 27 locations and 160 dates, same known/held-out role counts. Matching uses pretest log daily gross demand and PV capacity; standardized differences are approximately -0.029 and -0.011 . The dispersed cohort spans 21 postcodes with no more than two households per postcode. Units follow outcome names: AUD/owner-day, percentage points, kW or kWh/day. No real-neighborhood connectivity or causal geographic effect is inferred.

The main nine deployment cells retain three-seed paired household-level cost and voltage summaries. Among full ToU owners, 3.6–7.3% have higher mean cost across fixed-panel/adoption combinations. In the 28-owner case, nonowner exposure changes are $-0.3479/-0.2793/-0.2725$ percentage points across the three fixed panels, versus $-0.5317/-0.5103/-0.4881$ for owners. Both groups can benefit on average while individual sites or customers still lose. Capacity and pretest-demand tertile cutpoints are fixed from known pretest records, not selected for subgroup significance; the full results remain descriptive without additional subgroup significance claims.

S10. External data and freeze chronology

Table S10. All external scale, tariff and seed outcomes.

Reference	Tariff	Training seed	Cost Δ (cents/day)	95% interval
2.2	tou	11	-1.942	-2.457 to -1.448
2.2	tou	23	-1.412	-2.012 to -0.794
2.2	tou	47	-0.400	-0.816 to 0.014
2.2	tou	Seed mean	-1.252	-1.741 to -0.767
2.2	flat	11	-0.383	-0.706 to -0.070
2.2	flat	23	-0.107	-0.381 to 0.172
2.2	flat	47	0.857	0.431 to 1.283
2.2	flat	Seed mean	0.122	-0.194 to 0.458

Reference	Tariff	Training seed	Cost Δ (cents/day)	95% interval
2.0	tou	11	-2.140	-2.578 to -1.707
2.0	tou	23	-1.759	-2.289 to -1.237
2.0	tou	47	-0.818	-1.220 to -0.431
2.0	tou	Seed mean	-1.573	-2.017 to -1.139
2.0	flat	11	-0.276	-0.588 to 0.034
2.0	flat	23	-0.072	-0.323 to 0.183
2.0	flat	47	0.767	0.365 to 1.170
2.0	flat	Seed mean	0.140	-0.165 to 0.444
2.4	tou	11	-1.628	-2.175 to -1.115
2.4	tou	23	-0.922	-1.587 to -0.236
2.4	tou	47	0.140	-0.299 to 0.551
2.4	tou	Seed mean	-0.803	-1.332 to -0.275
2.4	flat	11	-0.496	-0.850 to -0.157
2.4	flat	23	-0.191	-0.470 to 0.100
2.4	flat	47	0.914	0.480 to 1.357
2.4	flat	Seed mean	0.076	-0.261 to 0.415

Each row uses 2,224 identical scored household-days across ten verified PV households in 2020; all references are normalization scales rather than verified nameplates. Negative Δ favors DFL. Three-seed rows average paired daily differences before resampling all 366 calendar positions. Flat-price checkpoints carry hashes and original timestamps verified after evaluation.

All 20 StoreNet power CSV files were downloaded and inspected. Verified PV household IDs are 1,2,3,4,5,7,10,11,13,17. There are 3,439 target days passing interval quality before the seven-day-history requirement and 2,224 scored days after it. Mean nominal capacity is not inferred from production maxima. A half-hour must contain at least 27 of 30 valid minutes in each required channel; available power observations are averaged without interpolation. Original minute-grid power missingness is retained; the companion interpolated energy pipeline is unused. Public OPSP household data were also assessed, but cumulative-meter interpretation, load reconstruction and insufficient verified capacity fields made them a weaker frozen-transfer candidate; its download and screening record is retained without claiming an external result from that dataset.

S11. Numerical and independent consistency checks

Original strictly convex battery-layer checks include primal state and energy balance, KKT differentiation, action selection under alternative regularization, independent settlement, and an equal-buy/sell negative control. The reported checks add independent measured-cost recomputation on the expanded dispatch, a real-data continuous-SOC gradient audit, inventory-boundary reconciliation and per-element export-curtailment and lost-revenue verification. AC local sensitivities are checked against full nonlinear replay and halved perturbations. These checks distinguish model implementation from field fidelity; no solver residual is treated as evidence about unknown household wiring, impedances, phase assignments, or inverter response.

State and settlement checks passed for all 78 completed dispatch files, comprising 42 Ausgrid and 36 external files. External recomputation had zero stored-cost discrepancy; the largest rounded-state bound and cyclic-end discrepancies were 1.43e-06 and 1.91e-06 kWh. An additional independent settlement implementation recomputed all 18 expanded comparison files directly from observed load, generation and saved charge/discharge actions, with no failed or unfinished file in that comparison.

The network analysis contains 133 completed scenario results: 130 nonlinear AC replays and 3 geographic replays restricted to the common-date mask. All new replays converged, and the maximum absolute active-power balance residual was $1.08\text{e-}05$ kW. Numerical convergence does not imply satisfaction of voltage or equipment constraints. Export checks include curtailed generation and foregone export revenue.

The local reproduction package contains source-data acquisition records, model weights, derived inputs, saved dispatch, numerical audits and the code required to reconstruct the reported tables and figures. It supports independent cost, observation-mask and checkpoint checks. Fresh training from source data additionally requires the documented public-data acquisition and cache-generation steps. Source-data and software licenses are recorded with the corresponding materials; the package is a local research artifact and has not been assigned a public DOI.